

# identification of seakeeping models from freq-response data with structure and parameter constraints

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In 2007, we started a new  
programme:

# motivation for this seminar

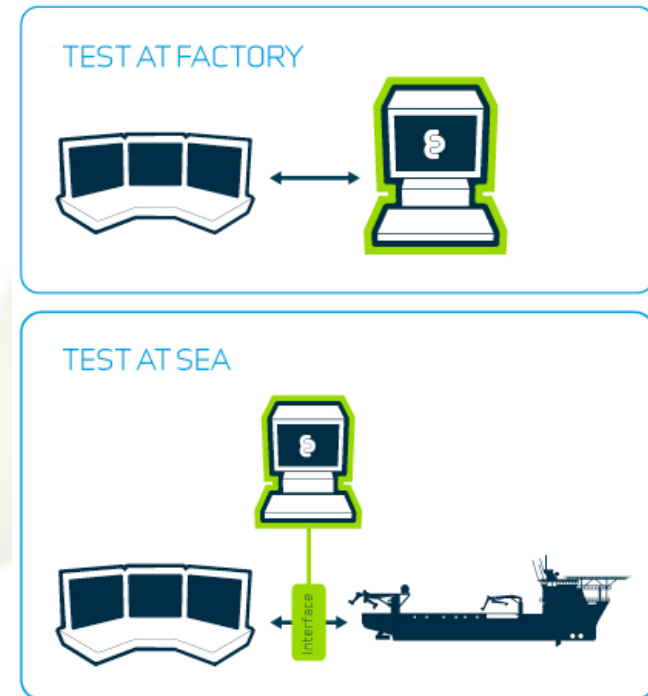
Linear time-domain models of marine structures are the basis of

- Training simulators.
- Hardware-in-the-loop (HIL) simulators.
- Motion control system design.
- Model-based fault detection and diagnosis.

Improvements in accuracy and speed can be achieved by appropriate model representations.



## HIL Simulator



# motivation for this seminar

One simple way to obtain such models is to use potential theory codes to compute **non-parametric models** hydrodynamic (coefficients and frequency responses) and then

- make a direct implementation of the Cummins Equation, or
- use system identification to approximate the Cummins equation by a Linear-time-invariant **parametric models**.

Approximations by LTI models in terms of state-space equations result in much faster simulations.

Depending on the complexity of the model simulations can be up to 80 times faster (Taghipour, Perez, and Moan, 2008).

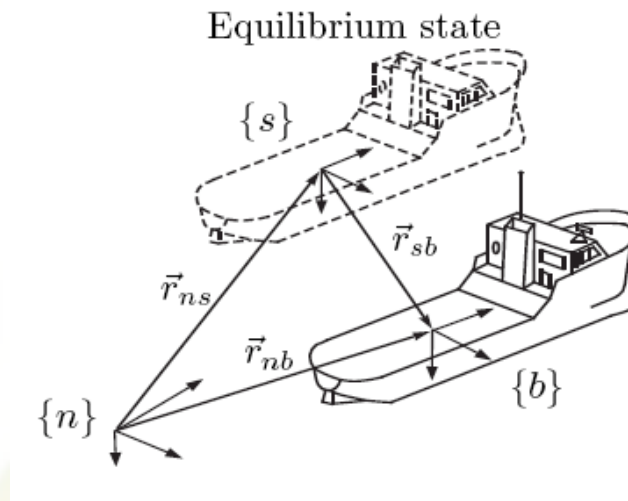
# outline for this presentation

- linear dynamic models of marine structures
  - equations of motion
  - Cummins equation
  - non-parametric models
  - model properties derived from hydrodynamics
- parametric approximations
  - consequence of the properties
  - time-domain Identification
  - frequency-domain identification
  - dealing with 2D hydrodynamic data
- examples: container, semi-submersible, fpso
- Discussion

# equations of motion

## Equations of Motion (rigid body):

$$\dot{\xi} = \mathbf{J}(\xi)\nu$$
$$\mathbf{M}_{RB}\dot{\nu} + \mathbf{C}_{RB}(\nu)\nu = \tau$$



$$\xi \triangleq [x, y, z, \phi, \theta, \psi]^T$$

← generalised displ. wrt equilibrium

$$\nu \triangleq [u, v, w, p, q, r]^T$$

← body-fixed generalised velocities

$$\tau \triangleq [X, Y, Z, K, M, N]^T$$

← body-fixed generalised forces

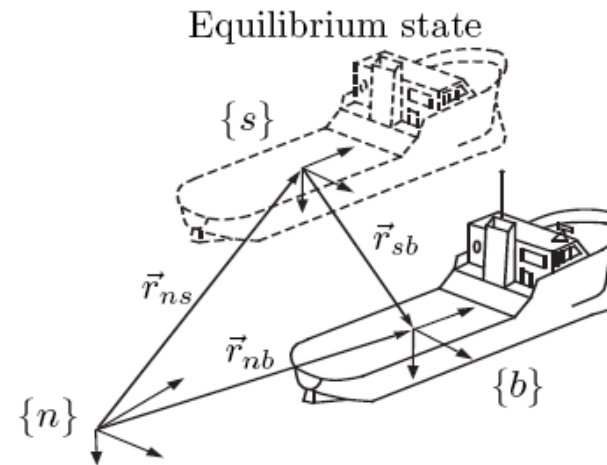
# linear equations

If we consider small deviations about the equilibrium:

$$\dot{\xi} = J(\xi)\nu$$
$$\mathbf{M}_{RB}\dot{\nu} + \mathbf{C}_{RB}(\nu)\nu = \tau$$



$$\mathbf{M}_{RB}\ddot{\xi} = \tau$$



Superposition of forces (radiation, restoring excitation):

$$\tau = \tau_{rad} + \tau_{res} + \tau_{exc}$$

# Cummins's equation

Cummins studied the radiation potential problem in the time-domain, within the linear assumption and found that:

$$\tau_{rad} = -\mathbf{A}_{\infty} \ddot{\xi} - \int_0^t \mathbf{K}(t - t') \dot{\xi}(t') dt'$$

Combining terms,

$$(\mathbf{M}_{RB} + \mathbf{A}_{\infty}) \ddot{\xi} + \int_0^t \mathbf{K}(t - t') \dot{\xi}(t') dt' + \mathbf{G}\xi = \tau_{exc}$$

This equation form the basis of most time-domain simulators.

# Ogilvie's relations

When considered in the frequency-domain, the radiation forces and the response of the marine structure can be expressed as:

$$\tau_{rad}(j\omega) = -\mathbf{A}(\omega)\ddot{\boldsymbol{\xi}}(j\omega) - \mathbf{B}(\omega)\dot{\boldsymbol{\xi}}(j\omega)$$

$$[-\omega^2[\mathbf{M} + \mathbf{A}(\omega)] + j\omega\mathbf{B}(\omega) + \mathbf{G}]\boldsymbol{\xi}(j\omega) = \tau_{exc}(j\omega)$$

Ogilvie found the relation between the Cummins parameters and the above:

$$\mathbf{A}(\omega) = \mathbf{A}_{\infty} - \frac{1}{\omega} \int_0^{\infty} \mathbf{K}(t) \sin(\omega t) dt.$$
$$\mathbf{B}(\omega) = \int_0^{\infty} \mathbf{K}(t) \cos(\omega t) dt.$$
$$\mathbf{A}_{\infty} = \lim_{\omega \rightarrow \infty} \mathbf{A}(\omega)$$

# non-parametric models

Time-domain:

$$\mathbf{K}(t) = \frac{2}{\pi} \int_0^\infty \mathbf{B}(\omega) \cos(\omega t) d\omega.$$

Frequency-domain:

$$\mathbf{K}(j\omega) = \int_0^\infty \mathbf{K}(t) e^{-j\omega t} dt = \mathbf{B}(\omega) + j\omega[\mathbf{A}(\omega) - \mathbf{A}_\infty]$$

These relationships are key since they are the starting point for the identification process by which parametric models are obtained.

# convolution replacement

The convolution term is inconvenient to implement simulation tools, and also to analyse and design motion control systems.

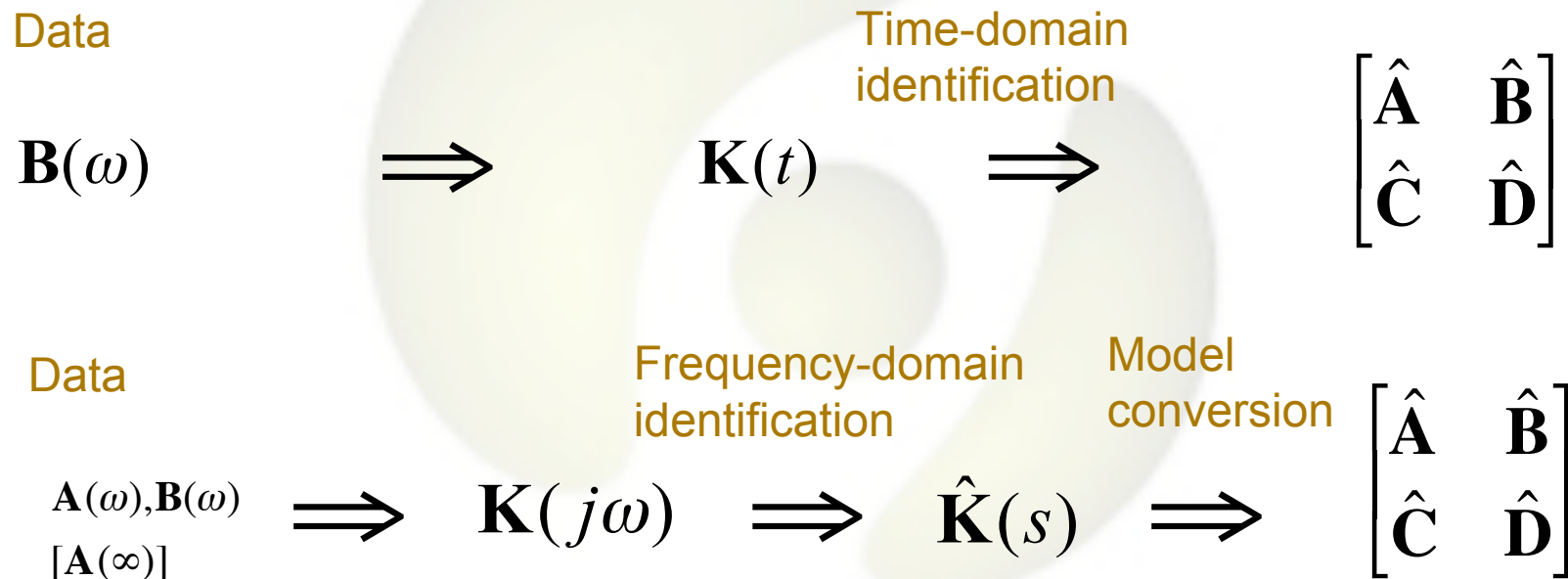
It is more convenient to seek a replacement by a state-space model:

$$\mu = \int_0^t \mathbf{K}(t - t') \dot{\xi}(t') dt' \quad \approx \quad \begin{aligned} \dot{\mathbf{x}} &= \hat{\mathbf{A}} \mathbf{x} + \hat{\mathbf{B}} \dot{\xi} \\ \hat{\mu} &= \hat{\mathbf{C}} \mathbf{x} + \hat{\mathbf{D}} \dot{\xi} \end{aligned}$$

- Different methods have been reported in the literature over the past 20 years.
- Due to Markovian properties of the SS-model significant gains in simulation speed can be obtained.

# convolution replacement

The convolution replacement can be posed in different ways:



In practice one method can be more favourable than the other.

# identification

Different proposals have appeared in the literature:

## Time-domain identification:

- LS-fitting of the impulse response (Yu & Falnes, 1998)
- Realization theory (Kristiansen & Egeland, 2003)

## Frequency-domain identification:

- LS-fitting of the frequency response (Jeffreys, 1984), (Damaren 2000).
- LS-fitting of added mass and damping (Soding 1982), (Xia et. al 1998), (Sutulo & Guedes-Soares 2006).

# properties of retardation functions

The following properties derive from the hydrodynamics, and have implications on the parametric models:

$$\hat{K}_{ik}(s) = \frac{P_{ik}(s)}{Q_{ik}(s)} = \frac{p_r s^r + p_{r-1} s^{r-1} + \dots + p_0}{s^n + q_{n-1} s^{n-1} + \dots + q_0}$$

| Property                                                      | Implication on Parametric Models $K_{ik}(s) = P(s)/Q(s)$                                  |
|---------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| 1) $\lim_{\omega \rightarrow 0} \mathbf{K}(j\omega) = 0$      | There are zeros at $s = 0$ .                                                              |
| 2) $\lim_{\omega \rightarrow \infty} \mathbf{K}(j\omega) = 0$ | Strictly proper.                                                                          |
| 3) $\lim_{t \rightarrow 0^+} \mathbf{K}(t) \neq 0$            | Relative degree 1.                                                                        |
| 4) $\lim_{t \rightarrow \infty} \mathbf{K}(t) = 0$            | BIBO stable.                                                                              |
| 5) The mapping $\dot{\xi} \mapsto \mu$ is Passive             | $\mathbf{K}(j\omega)$ is positive real (diagonal entries $K_{ii}(j\omega)$ positive real. |

This prior knowledge, and should be used in the identification process to refine the search for approximating models.

# time-domain methods

identification from the impulse-response

# time-domain methods

- Impulse response curve fitting (Yu & Falnes, 1995, 1998)
- Realization theory (Kristiansen & Egeland, 2003)

Both these methods use the frequency domain data to compute the retardation functions in the time domain and then perform the system identification.

$$\mathbf{K}(t) = \frac{2}{\pi} \int_0^{\infty} \mathbf{B}(\omega) \cos(\omega t) d\omega.$$

# distortion of non-param. models

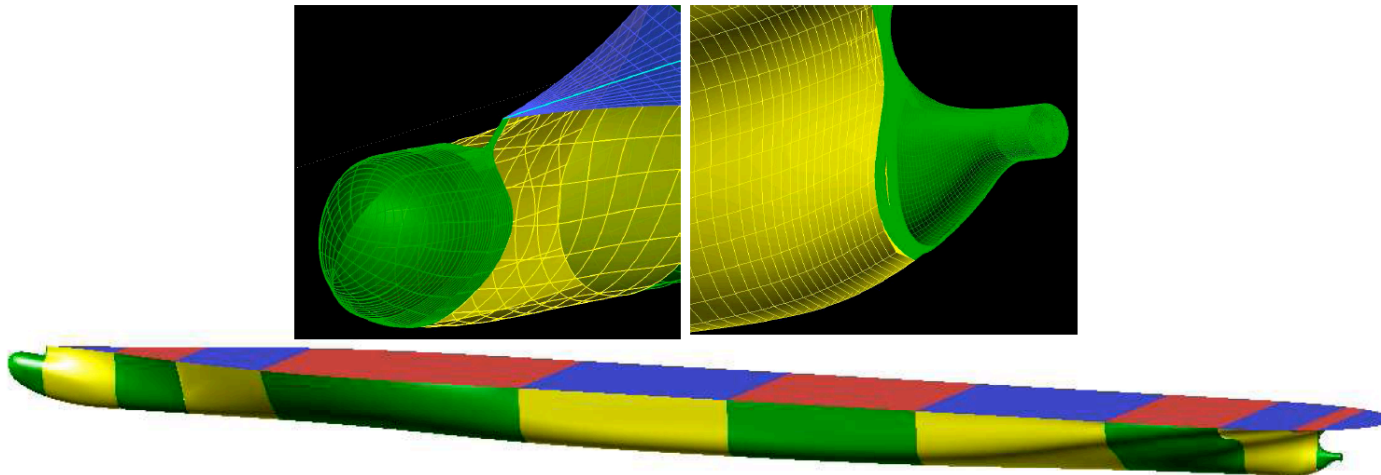
The computation of the retardation function from the damping introduces **distortion**, which can affect the identification:

$$\mathbf{K}(t) \approx \bar{\mathbf{K}}(t) = \frac{2}{\pi} \int_0^{\Omega} \mathbf{B}(\omega) \cos(\omega t) d\omega$$

The maximum frequency is related to the size of the panels. One can use asymptotic tails to extend the computations:

$$\text{as } \omega \rightarrow \infty, \quad B_{ik}(\omega) \rightarrow \frac{\beta_1}{\omega^4} + \frac{\beta_2}{\omega^2}$$

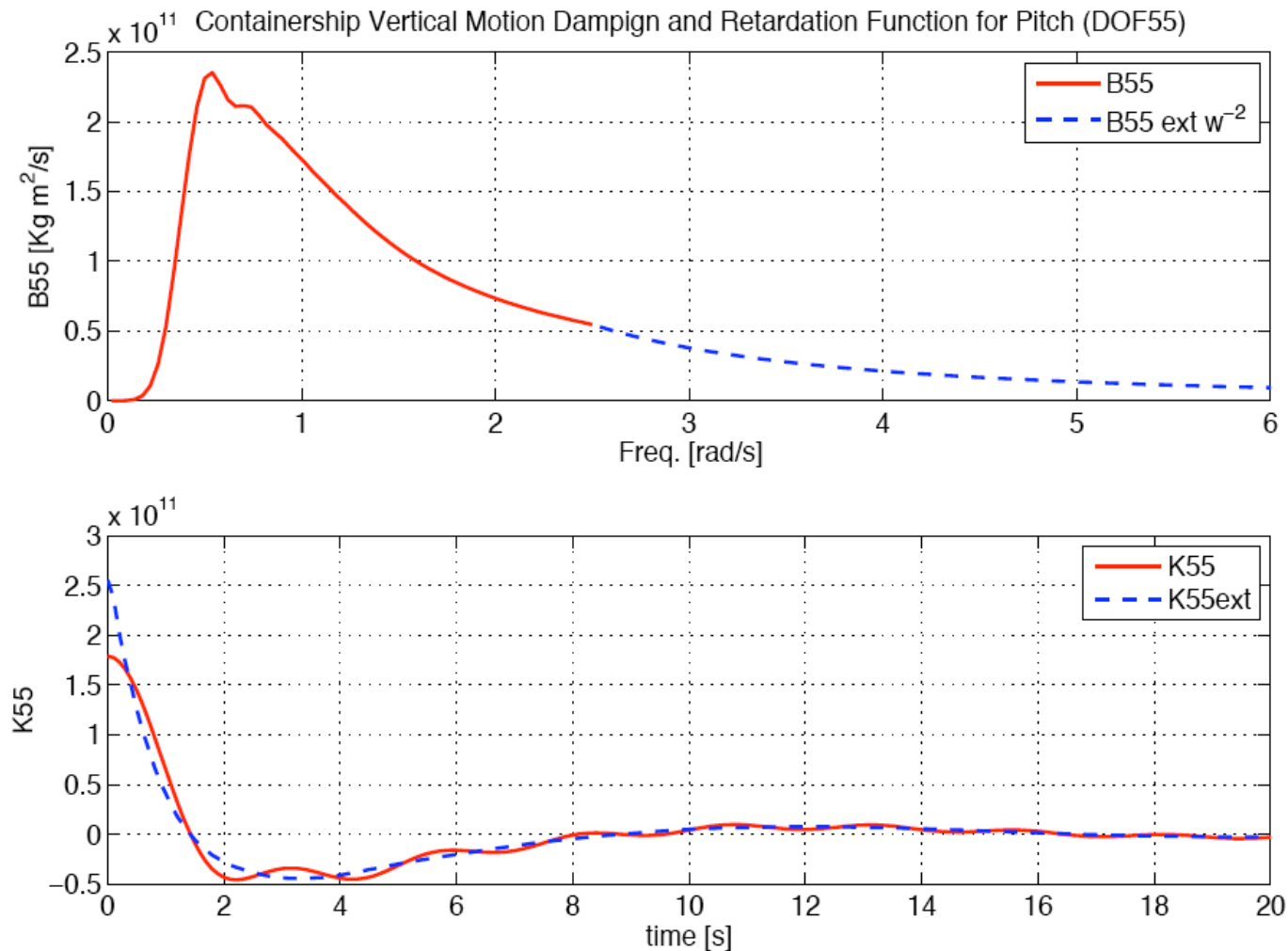
# example container ship (Taghipour et al., 2007a)



The panel sizing was done to be able to compute frequencies up to 2.5 rad/s.

From experience characteristic panel length  $< 1/8$ - $1/10$  min wave length for low order panel methods (Faltinsen, 1993).

# distortion of non-param. models



# distortion of non-param. models

We can think the limited freq. damping as a product of the real damping with a rectangular window:

$$\bar{\mathbf{K}}(t) = \frac{2}{\pi} \int_0^{\infty} \mathbf{W}(\omega) \mathbf{B}(\omega) \cos(\omega t) d\omega \quad W_{ik}(\omega) = \begin{cases} 1 & \text{if } \omega \leq \Omega, \\ 0 & \text{if } \omega > \Omega. \end{cases}$$

Product in the freq.-domain  $\Leftrightarrow$  convolution in the time-domain (FT property):

$$\bar{\mathbf{K}}(t) = \int_0^{\infty} \mathbf{W}(t - t') \mathbf{K}(t) dt' \quad W_{ik}(t) = 2\Omega \frac{\sin(\Omega t)}{\Omega t}$$

The distortion can be expressed as the convolution of the true impulse response and the IFT of an ideal low-pass filter.

This is a disadvantage for time-domain methods.

# impulse response curve fitting

$$\theta_{ik}^* = \arg \min_{\theta} \sum_l [\bar{K}_{ik}(t_l) - \hat{K}_{ik}(t_l, \theta)]^2$$

$$\hat{K}_{ik}(t, \theta) = \hat{C}_{ik}(\theta) \exp(\hat{A}_{ik}(\theta)t) \hat{B}_{ik}(\theta) + \hat{D}_{ik}(\theta)$$

- Optimisation problem non-linear in the parameters. Algorithms can be trapped in local minima. A good guess of the initial parameters is crucial for the success of the optimisation.
- The structure of the state-space model adopted plays an important role--there are infinite ways of doing this.
- The order of the model and the initial parameters are not easy to guess from the impulse response.
- Make no use of the prior knowledge.

# realization theory

Discrete-time approximation:

$$\begin{aligned} \mathbf{x}_{k+1} &= \Phi \mathbf{x}_k + \Gamma u_k \\ y_k &= \mathbf{C} \mathbf{x}_k + \mathbf{D} u_k \end{aligned} \quad \Rightarrow \quad K_k = \mathbf{C} \Phi^{k-1} \Gamma + \mathbf{D}$$

Steps:

- 1) Form a Hankel matrix with the impulse response samples.
- 2) Do a singular value decomposition (SVD).
- 3) Obtain the order from the number of non-zero singular values.
- 4) Obtain the model matrices via factorisation.
- 5) Convert the model to continuous time.

$$\mathcal{H}_k = \begin{bmatrix} K_1 & K_2 & \dots & K_k \\ K_2 & K_3 & \dots & K_{k+1} \\ \vdots & \vdots & & \vdots \\ K_k & K_{k+1} & \dots & K_{2k-1} \end{bmatrix}$$

$$\mathcal{H}_k = [\mathbf{U}_1 \mathbf{U}_2] \begin{bmatrix} \Sigma_1 & 0 \\ 0 & \Sigma_2 \end{bmatrix} [\mathbf{V}_1^* \mathbf{V}_2^*] = \mathbf{U}_1 \Sigma_1 \mathbf{V}_1^*$$

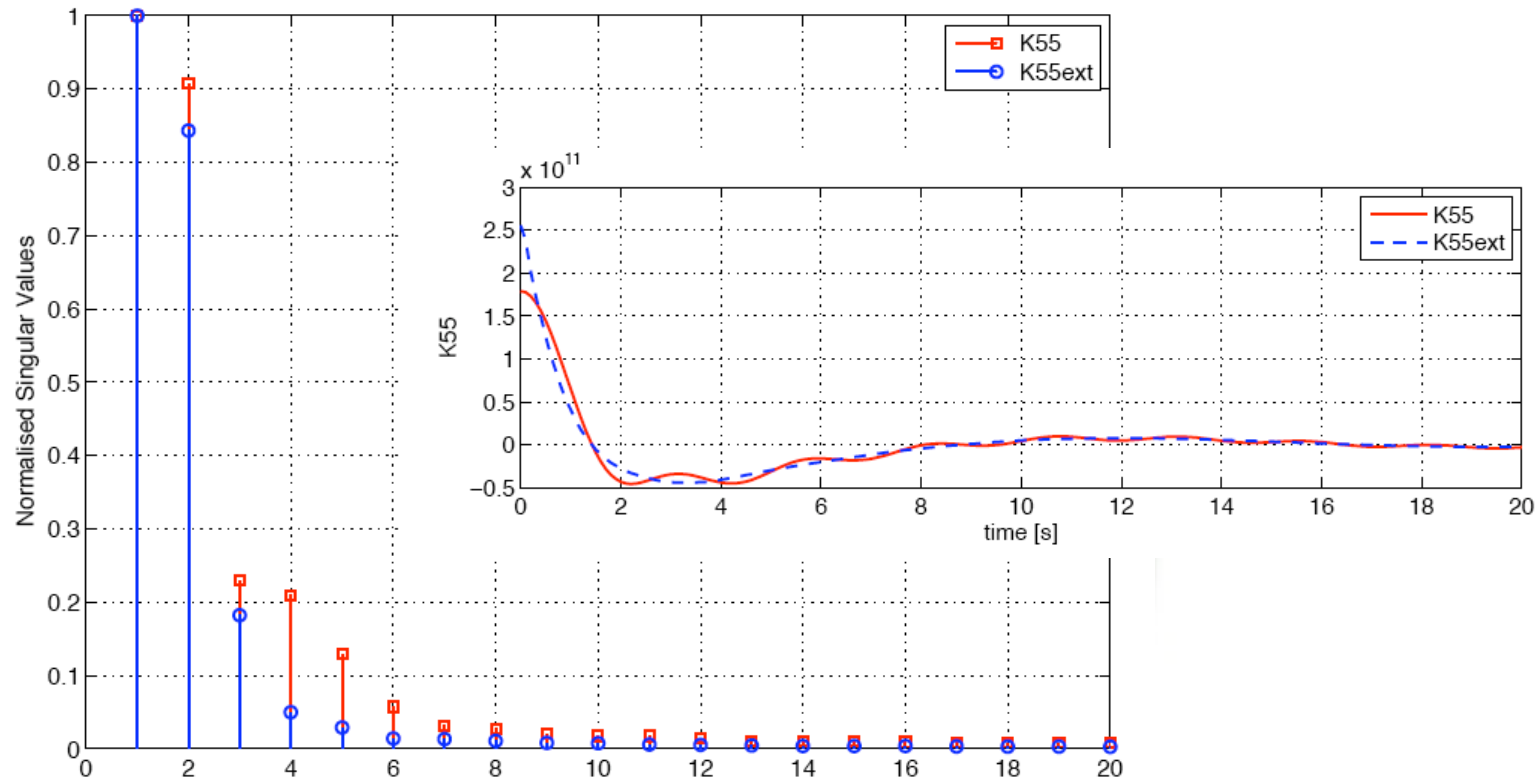
$$\Phi = \Sigma_1^{-1/2} \begin{bmatrix} \mathbf{U}_{11} \\ \mathbf{U}_{12} \end{bmatrix}^T \begin{bmatrix} \mathbf{U}_{12} \\ \mathbf{U}_{13} \end{bmatrix} \Sigma_1^{1/2}$$

$$\Gamma = \Sigma_1^{-1/2} \mathbf{V}_{11}^*$$

$$\mathbf{C} = \mathbf{U}_{11} \Sigma_1^{1/2}$$

$$\mathbf{D} = h(0),$$

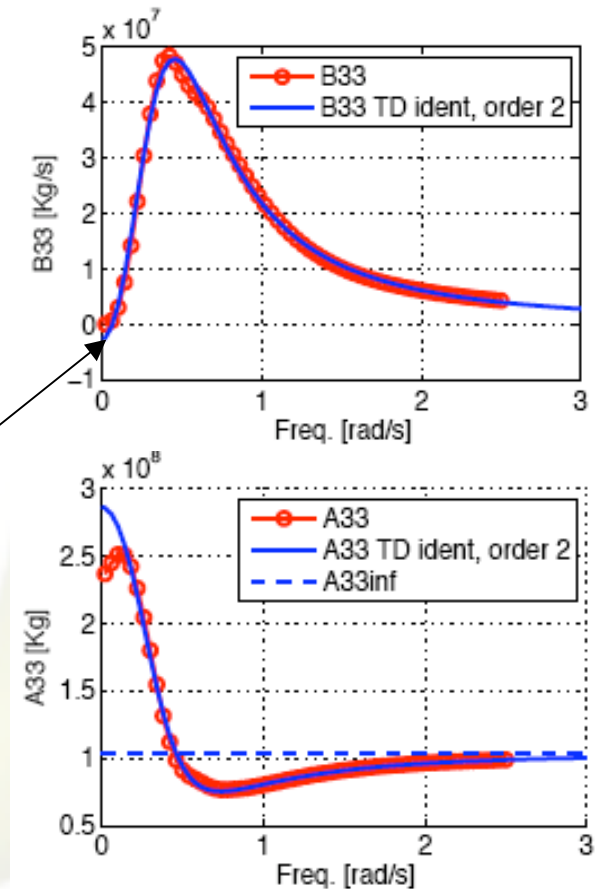
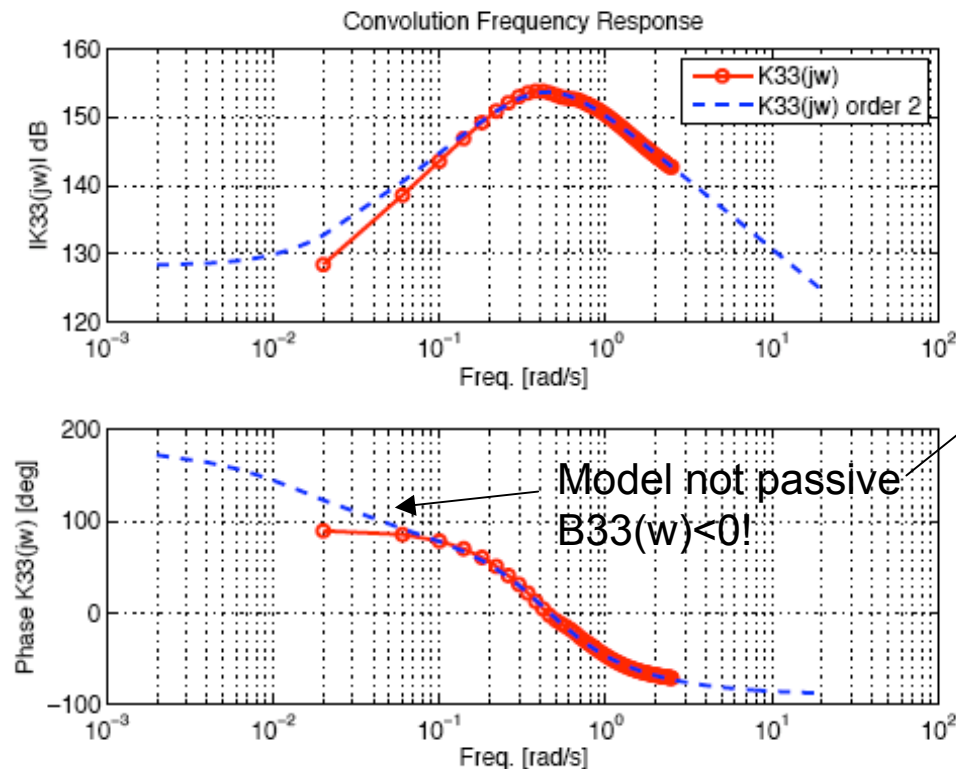
# order detection via SVD



The quality of the impulse response affects the order selection.

The containership example suggests K55(s) order 2 to 5

# example container ship (DOF33)



Reconstruction of damping and added mass from the parametric approximation:

$$\hat{A}(\omega) = \text{Im}\{\omega^{-1} \hat{K}(j\omega)\} + A_{\infty}$$

$$\hat{B}(\omega) = \text{Re}\{\hat{K}(j\omega)\},$$

# realization theory

- relatively easy to implement.
- do not require initial parameter estimates.
- allows order detection.
- requires conversion to continuous time (distortion).
- poor model quality.
- make no use of prior knowledge.

# frequency-domain methods

identification from the frequency-response

# frequency-response LS-fitting

The  $i,k$  entry of  $\mathbf{K}(s)$  can be approximated by a rational transfer function:

$$\hat{K}(s, \theta) = \frac{P(s, \theta)}{Q(s, \theta)} = \frac{p_m s^m + p_{m-1} s^{m-1} + \dots + p_0}{s^n + q_{n-1} s^{n-1} + \dots + q_0} \quad \theta = [p_m, \dots, p_0, q_{n-1}, \dots, q_0]^T$$

Then we can estimate the parameters via LS optimization using the frequency response computed using the data generated by hydrodynamic code:

$$\theta^* = \arg \min_{\theta} \sum_l w_l (\epsilon_l^* \epsilon_l) \quad \epsilon_l = K(j\omega_l) - \frac{P(j\omega_l, \theta)}{Q(j\omega_l, \theta)}$$

$$\mathbf{K}(j\omega) = \mathbf{B}(\omega) + j\omega[\mathbf{A}(\omega) - \mathbf{A}]$$

This problem is non-linear in the parameters, but it can be linearised.

# quasi-linear regression

for the non-linear problem

$$\theta^* = \arg \min_{\theta} \sum_l w_l (\epsilon_l^* \epsilon_l) \quad \epsilon_l = K(j\omega_l) - \frac{P(j\omega_l, \theta)}{Q(j\omega_l, \theta)}$$

Levi (1959) proposed a linearisation by choosing

$$\theta'^* = \arg \min_{\theta} \sum_l w'_l (\epsilon'_l{}^* \epsilon'_l), \quad \epsilon'_l = Q(j\omega_l, \theta) \epsilon_l = Q(j\omega_l, \theta) K(j\omega_l) - P(j\omega_l, \theta).$$

This is equivalent to choose the following weights in the original problem above:

$$w_l = w'_l |Q(j\omega_l, \theta)|^2$$

- This problem is linear in the parameters, a standard linear LS problem.
- It does not always give a good fit if data spans a large range of frequencies.

# iterative quasi-linear regression

Sanathanan and Koerner (1963), proposed an iterative solution via a sequence of linear LS problems:

1. Set  $\mathbf{W}_0 = \mathbf{I}$ .  $\epsilon' = [\epsilon'_1, \dots, \epsilon'_N]^T$
2. Solve  $\theta_k^* = \arg \min_{\theta} \epsilon'^* \mathbf{W}_k \epsilon'$ ,  $\mathbf{W} = \text{diag}(w'_1, w'_2, \dots, w'_n)$
3. Set  $\mathbf{W}_{k+1} = \text{diag}(|Q(j\omega_l, \theta_k)|^{-2})$  go to 2 until convergence.

This results in the following problem at each iteration  $k > 1$ :

$$\theta_k^* = \arg \min_{\theta} \sum_l \left| \frac{Q(j\omega_l, \theta)K(j\omega_l)}{Q(j\omega_l, \theta_{k-1}^*)} - \frac{P(j\omega_l, \theta)}{Q(j\omega_l, \theta_{k-1}^*)} \right|^2$$

After a few iterations  $\theta_k^* \approx \theta_{k-1}^*$  and we recover the original nonlinear problem.

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# prior knowledge: constraints

Adding prior knowledge is important to refine the search for models, and thus obtain better quality models.

Prior knowledge usually derives from the physics of the underlying problem; in this case, from the hydrodynamics.

# prior knowledge: constraints

For the transfer functions related to the convolution terms, we know

1. Relative degree 1
2.  $H_{ik}(s)=0$  for  $s=0$
3. Stable
4. Passive
5. Minimum order approximation is 2

Some of these properties can be enforced in the structure of the model and its parameters without complicating the optimisation.

# prior knowledge: constraints

Model:

$$\hat{K}_{ik}(s) = \frac{P_{ik}(s)}{Q_{ik}(s)} = \frac{p_r s^r + p_{r-1} s^{r-1} + \dots + p_0}{s^n + q_{n-1} s^{n-1} + \dots + q_0}, \quad i, k = 1, \dots, 6.$$

- Relative degree = 1  $\rightarrow$  Constrain  $r = n-1$
- Zero at  $s=0$

$$\begin{array}{l} P_{ik}(s) = s P'_{ik}(s) \\ \deg(P') = n - 2 \end{array} \quad \rightarrow \quad \hat{K}_{ik}(j\omega) = \frac{(j\omega) P'_{ik}(j\omega, \theta_{ik})}{Q_{ik}(j\omega, \theta_{ik})}$$

Redefine the problem:

$$\theta_{ik}^* = \arg \min_{\theta} \sum_l \left| \frac{K_{ik}(j\omega_l)}{(j\omega_l)} - \frac{P'_{ik}(j\omega_l, \theta)}{Q_{ik}(j\omega_l, \theta)} \right|^2$$

# prior knowledge: constraints

- **Stability:** The LS optimisation does not ensure stability, this is one way to force it after identification:

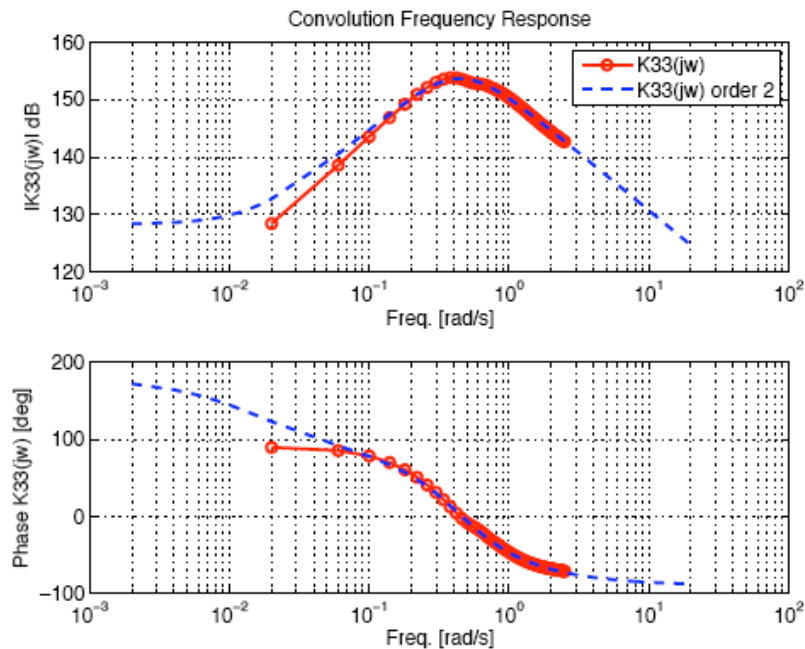
- (i) Compute the roots of  $\lambda_1, \dots, \lambda_n$  of  $Q_{ik}(s, \hat{\theta}_{ik})$ .
- (ii) If  $\text{Re}\{\lambda_i\} > 0$ , then set  $\text{Re}\{\lambda_i\} = -\text{Re}\{\lambda_i\}$ ,
- (iii) Reconstruct the polynomial:  $Q_{ik}(s) = (s - \lambda_1)(s - \lambda_1) \cdots (s - \lambda_n)$

- **Passivity:** Also not enforced by LS. Low-order models are usually passive, hence if not passive try reducing the order.
- **Minimum order:** The 2nd order approximation is the lowest order approximation that can satisfy all the properties of the retardation functions:

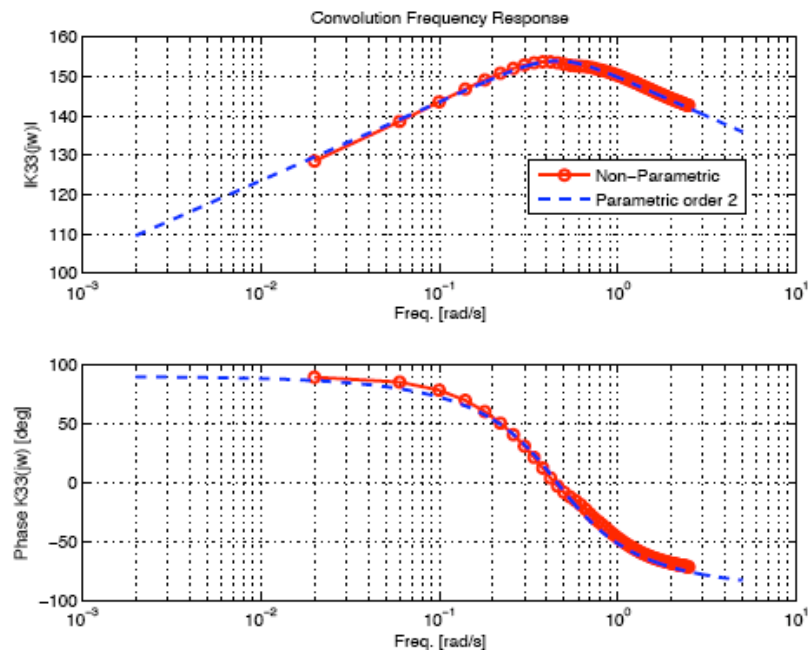
$$\hat{K}_{ik}^{min}(s) = \frac{p_0 s}{s^2 + q_1 s + q_0}$$

# example container (dof 33)

Realization Theory

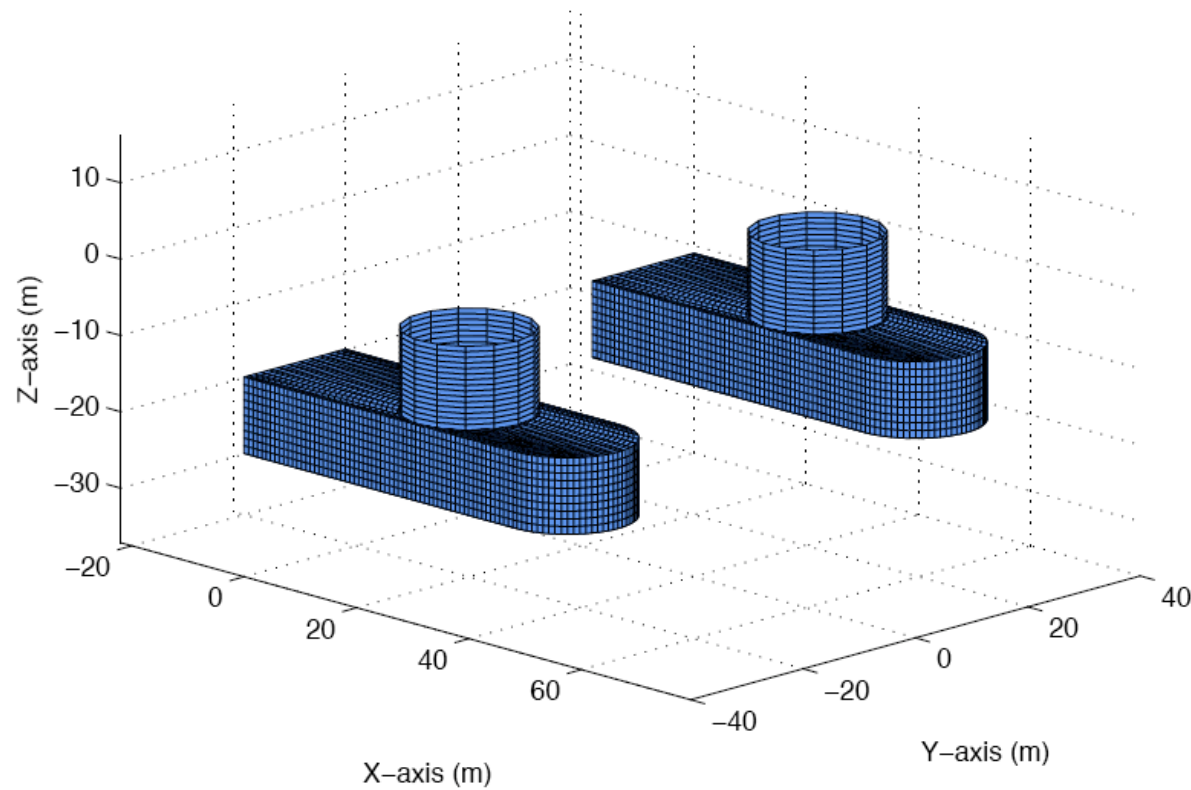


FD-Id with constraints



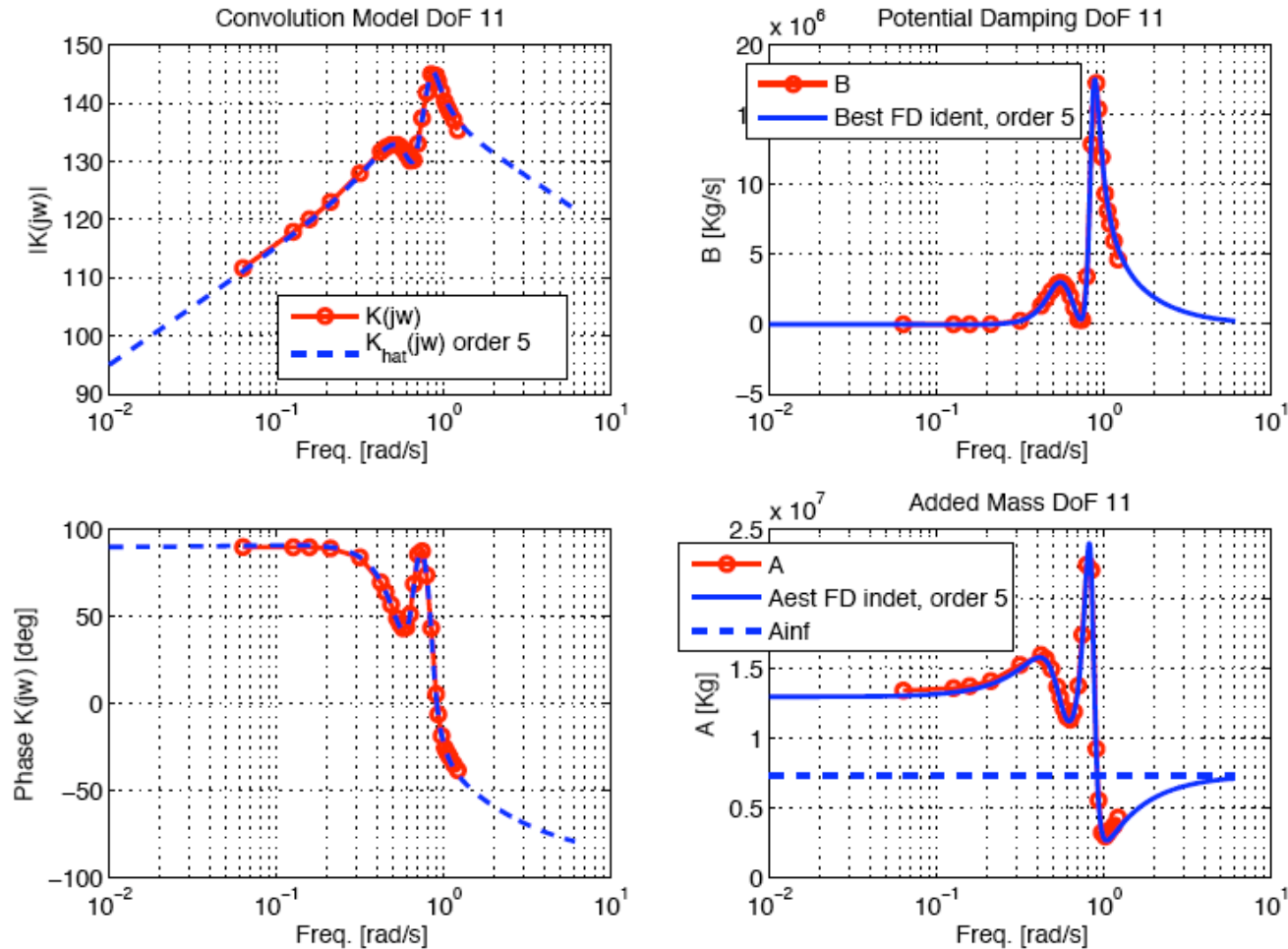
By imposing constraints on the model structure and parameters, we obtain a model that satisfy all the properties of the retardation functions and have a better quality.

# example semi-sub

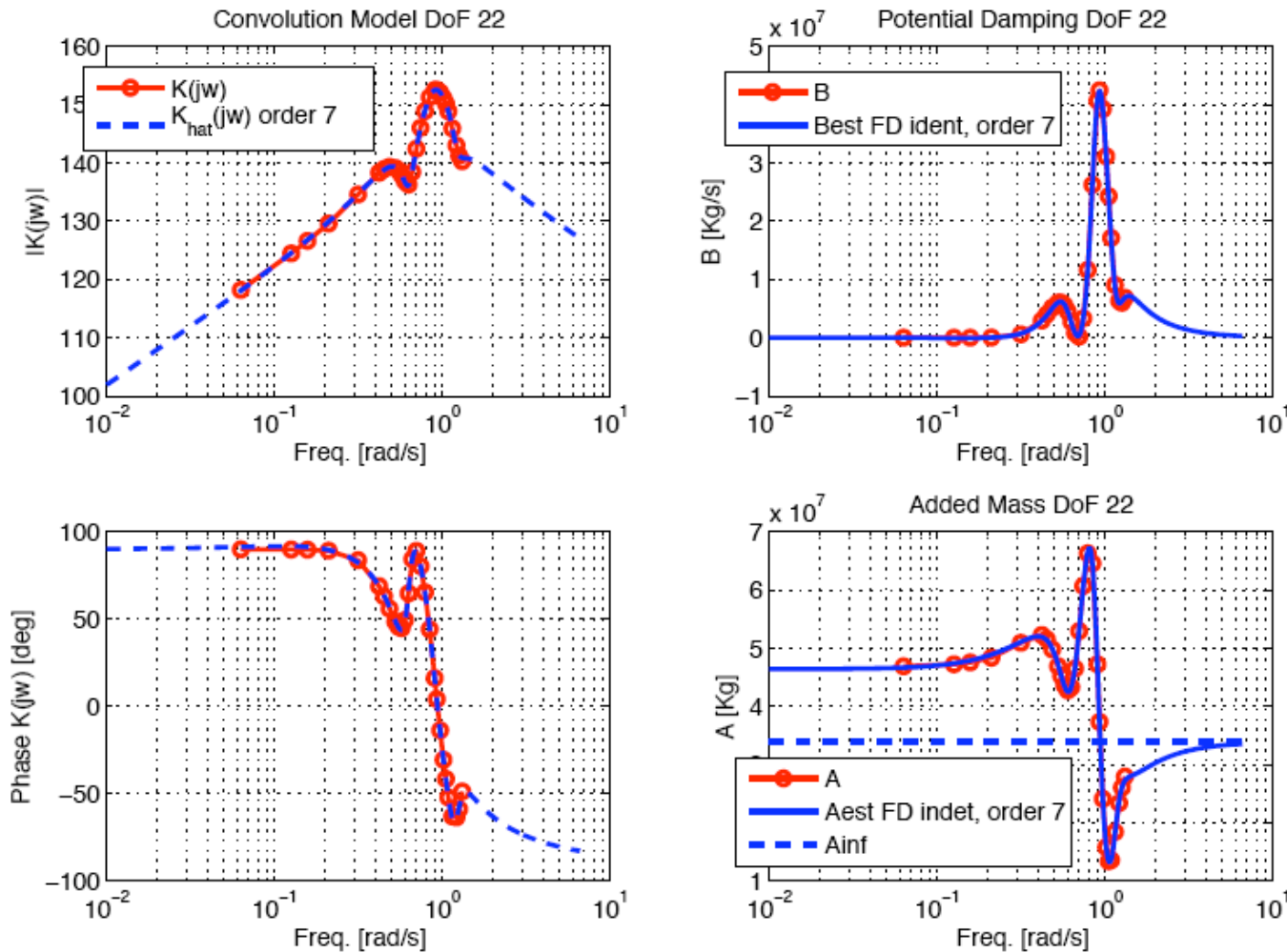


Data from [www.marinecontrol.org](http://www.marinecontrol.org)

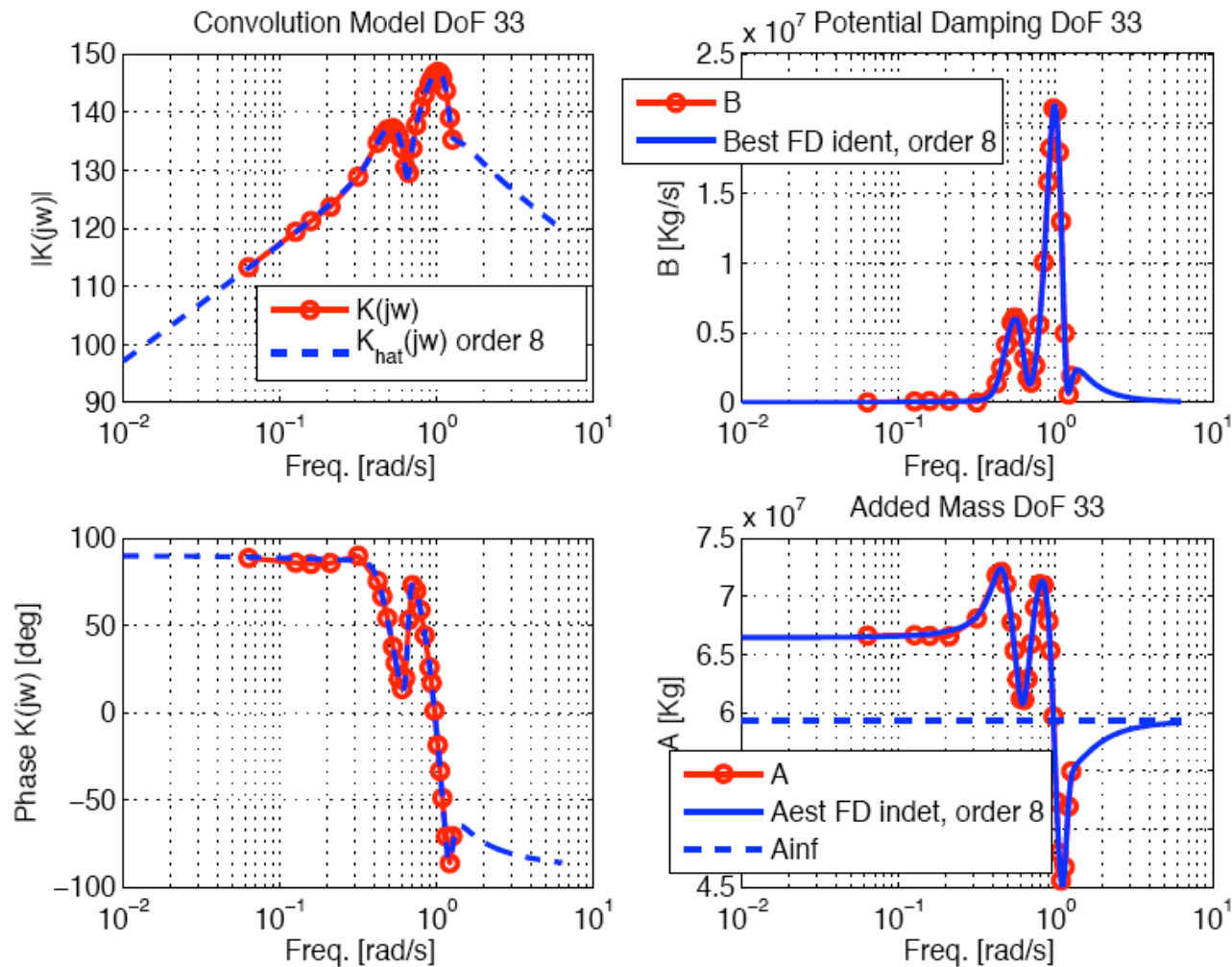
# example semi-sub: surge (ord. 5)



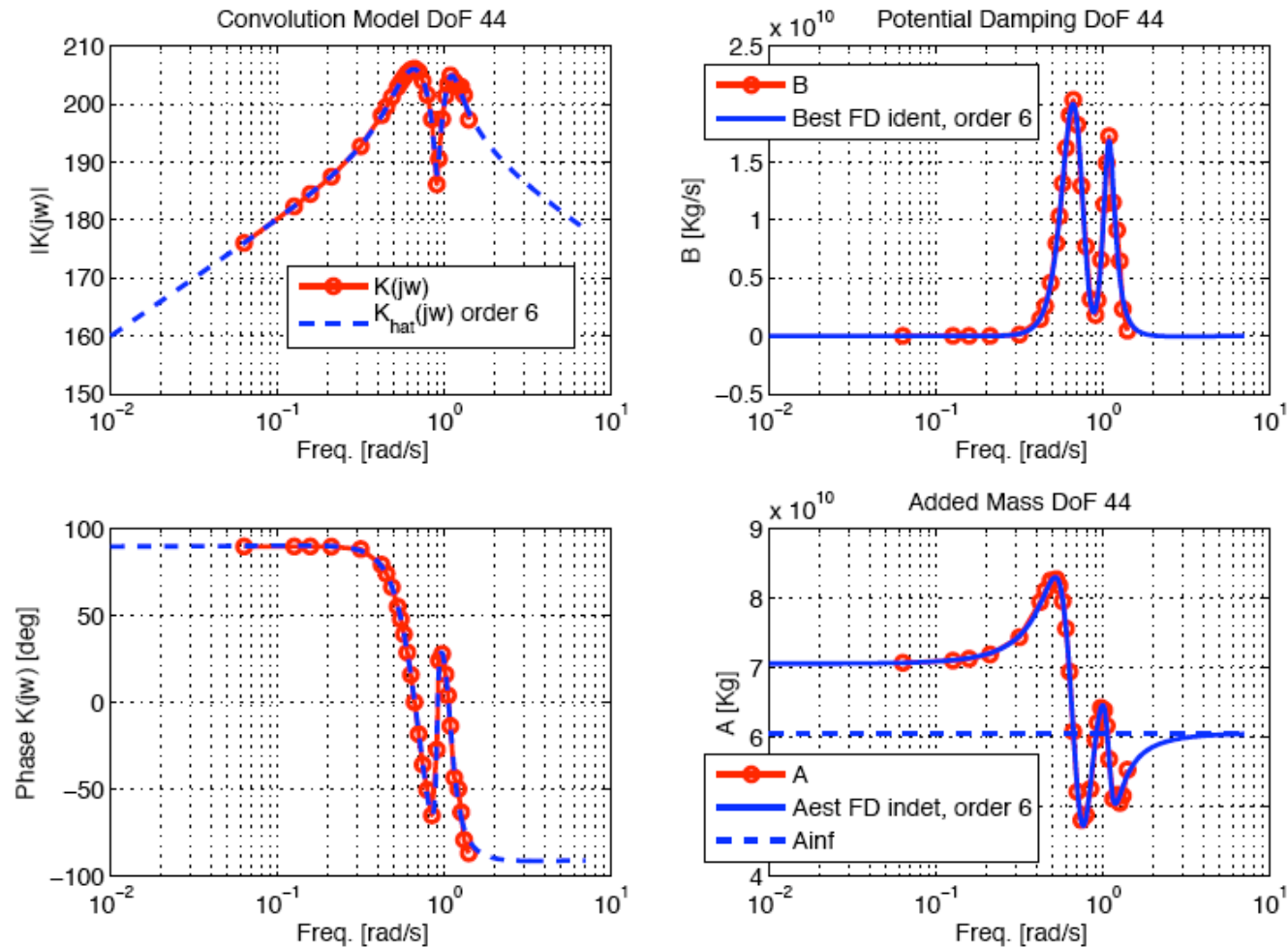
# example semi-sub: sway (ord. 7)



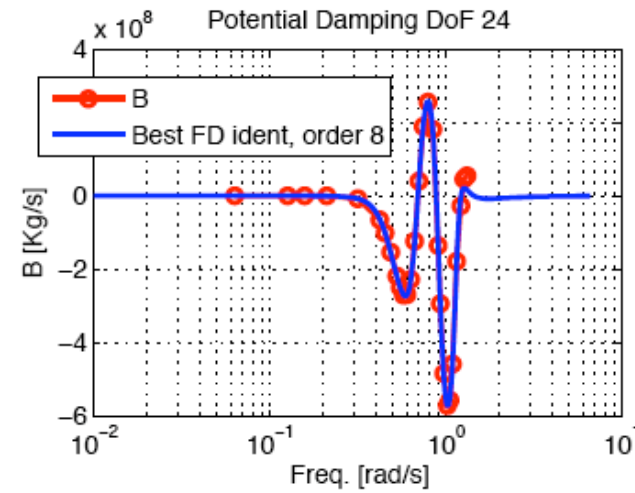
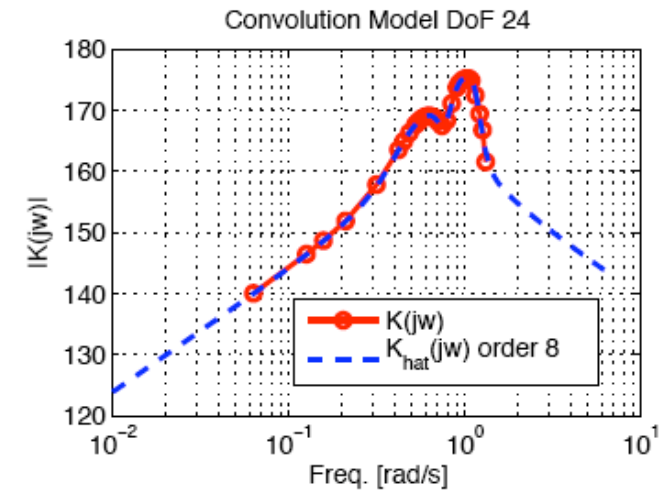
# example semi-sub: heave (ord. 8)



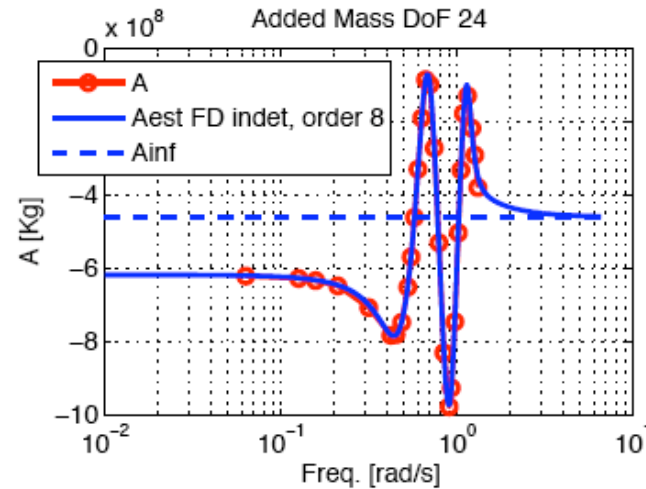
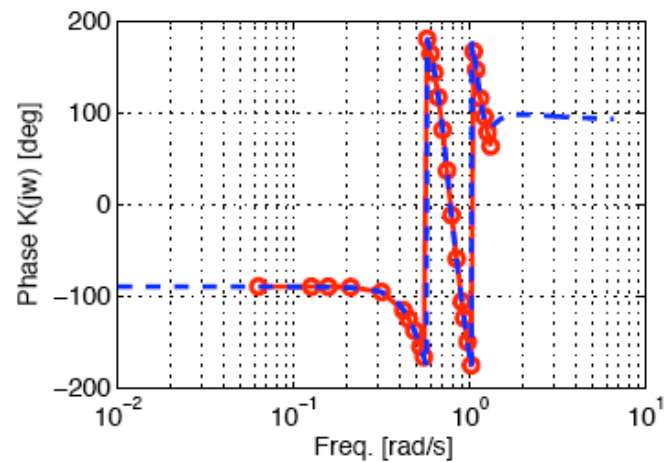
# example semi-sub: roll (ord. 6)



# example semi-sub: roll-sway (ord 8)



Couplings are not necessarily passive  $B(\omega) < 0$



# Frequency-domain methods

- Work directly in the FD, hence, there is no need to compute a distorted impulse response by going to time-domain.
- No need to compute data at very-low or very-high frequency. We can concentrate in the important range and compute many points.
- They can incorporate prior knowledge, and thus the models satisfy the physical characteristics of the problem. Good quality models.
- Order selection: one can start with the minimum order  $n=2$  and increase it if necessary to improve the fit. (check passivity.)
- Parameter estimation method is simple: a sequence of Linear Least-Squares problems.

# frequency-domain identification with 2D-data (strip-theory codes)

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joint identification of infinite-frequency  
added mass and memory models

# indirect FD identification

If we do not have access to  $\mathbf{A}^\infty$ , as in the case of strip theory codes, we can identify it together with the  $K(s)$ .

The non-parametric radiation force models in the frequency domain can be expressed as

$$\tau_{rad,ik}(j\omega) = \left[ \frac{B_{ik}(\omega)}{j\omega} + A_{ik}(\omega) \right] \ddot{\xi}(s)$$

From the parametric approximations, this can also be expressed as

$$\begin{aligned} \hat{\tau}_{rad,ik}(s) &= \left[ A_{\infty,ik} s + \frac{P_{ik}(s)}{Q_{ik}(s)} \right] \dot{\xi}(s), \\ &= \left[ A_{\infty,ik} + \frac{P'_{ik}(s)}{Q_{ik}(s)} \right] \ddot{\xi}(s), \end{aligned}$$

# indirect FD identification

Then we can define the complex coefficient:

$$\tilde{A}(j\omega) \triangleq \frac{B_{ik}(\omega)}{j\omega} + A_{ik}(\omega)$$

And fit to it a rational approximation

$$\theta^* = \arg \min_{\theta} \sum_l w_l (\epsilon_l^* \epsilon_l), \quad \epsilon_l = \tilde{A}_{ik}(j\omega_l) - \frac{R_{ik}(j\omega_l, \theta)}{S_{ik}(j\omega_l, \theta)}$$

with the constraint

$$n = \deg S_{ik}(s) = \deg R_{ik}(s) \quad \Leftrightarrow \quad \hat{\tilde{A}}_{ik}(s) = \frac{R_{ik}(s)}{S_{ik}(s)} = \frac{A_{\infty,ik} Q_{ik}(s) + P'_{ik}(s)}{Q_{ik}(s)}$$

# indirect FD identification

Once the  $R(s)$  and  $S(s)$  are obtained, we can obtain the added mass and fluid memory model from

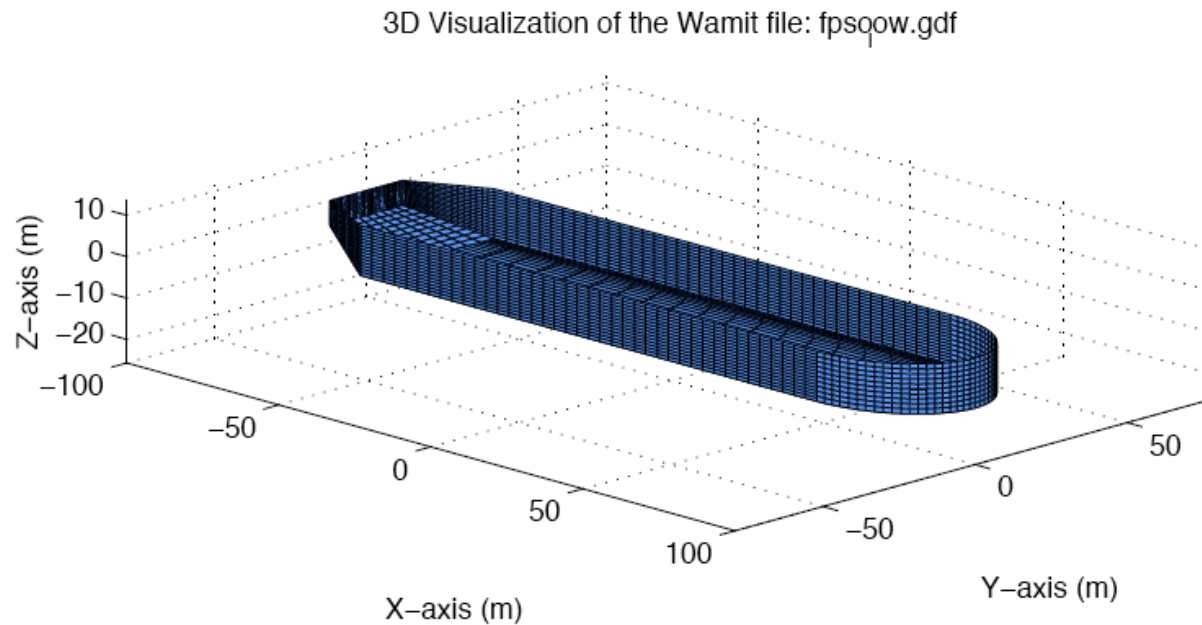
$$\hat{A}_{\infty,ik} = \lim_{\omega \rightarrow \infty} \frac{R_{ik}(s, \theta^*)}{S_{ik}(s, \theta^*)}$$

$$Q_{ik}(s, \theta^*) = S_{ik}(s, \theta^*),$$

$$P_{ik}(s, \theta^*) = R_{ik}(s, \theta^*) - \hat{A}_{\infty,ik} S_{ik}(s, \theta^*)$$

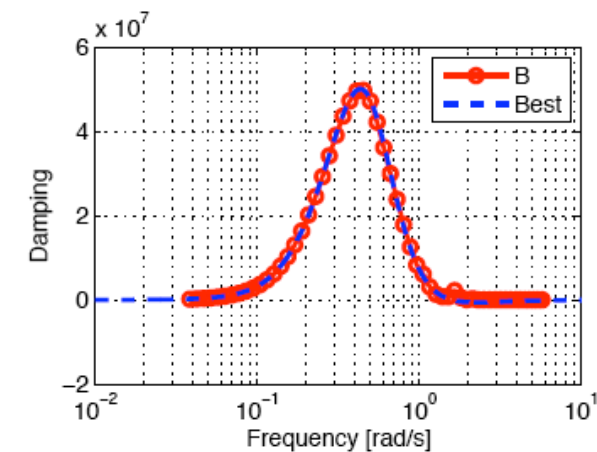
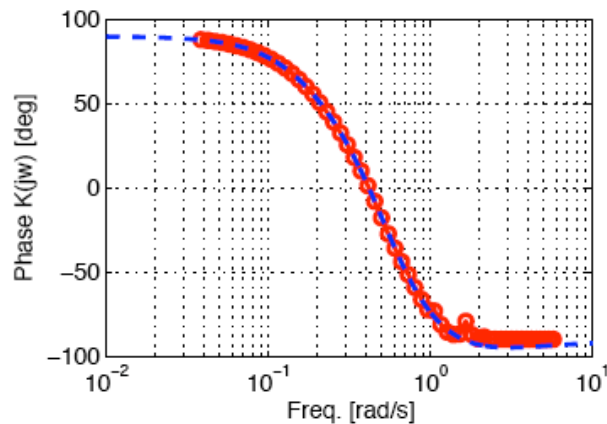
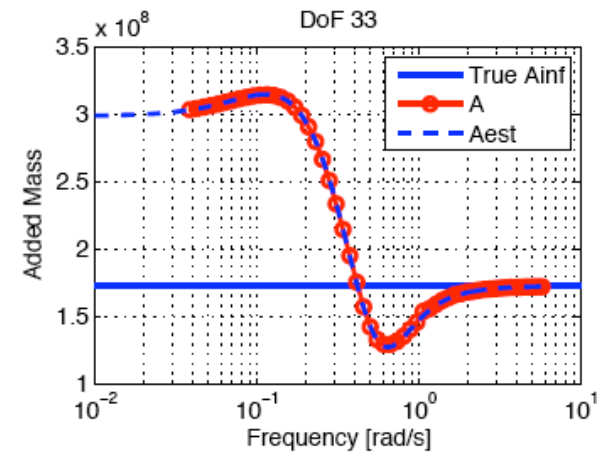
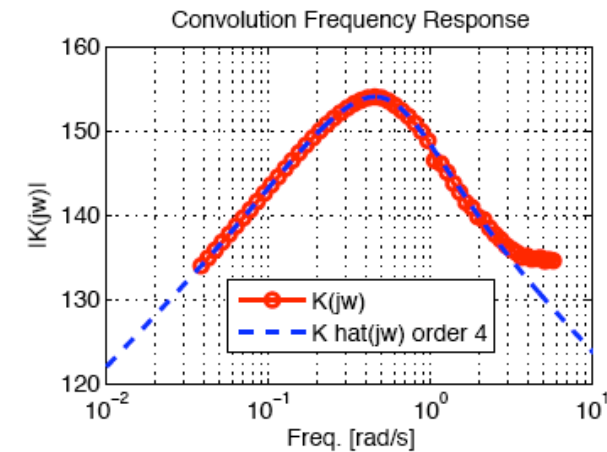
The coefficient  $\hat{A}_{\infty,ik}$  is the coefficient of the higher-order term of  $R_{ik}(s, \theta^*)$  if  $S_{ik}(s, \theta^*)$  is monic.

# example FPSO

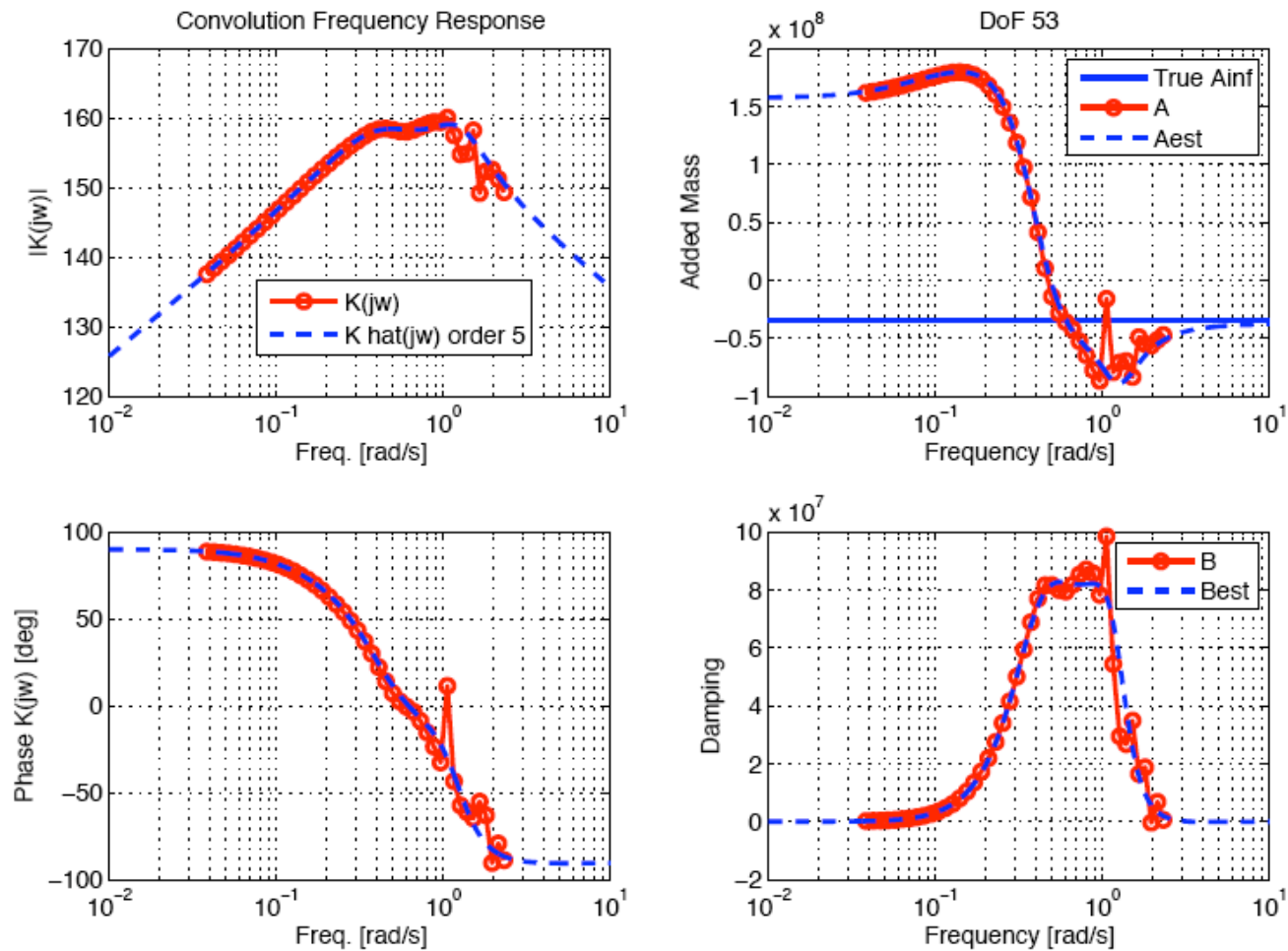


Data from [www.marinecontrol.org](http://www.marinecontrol.org)

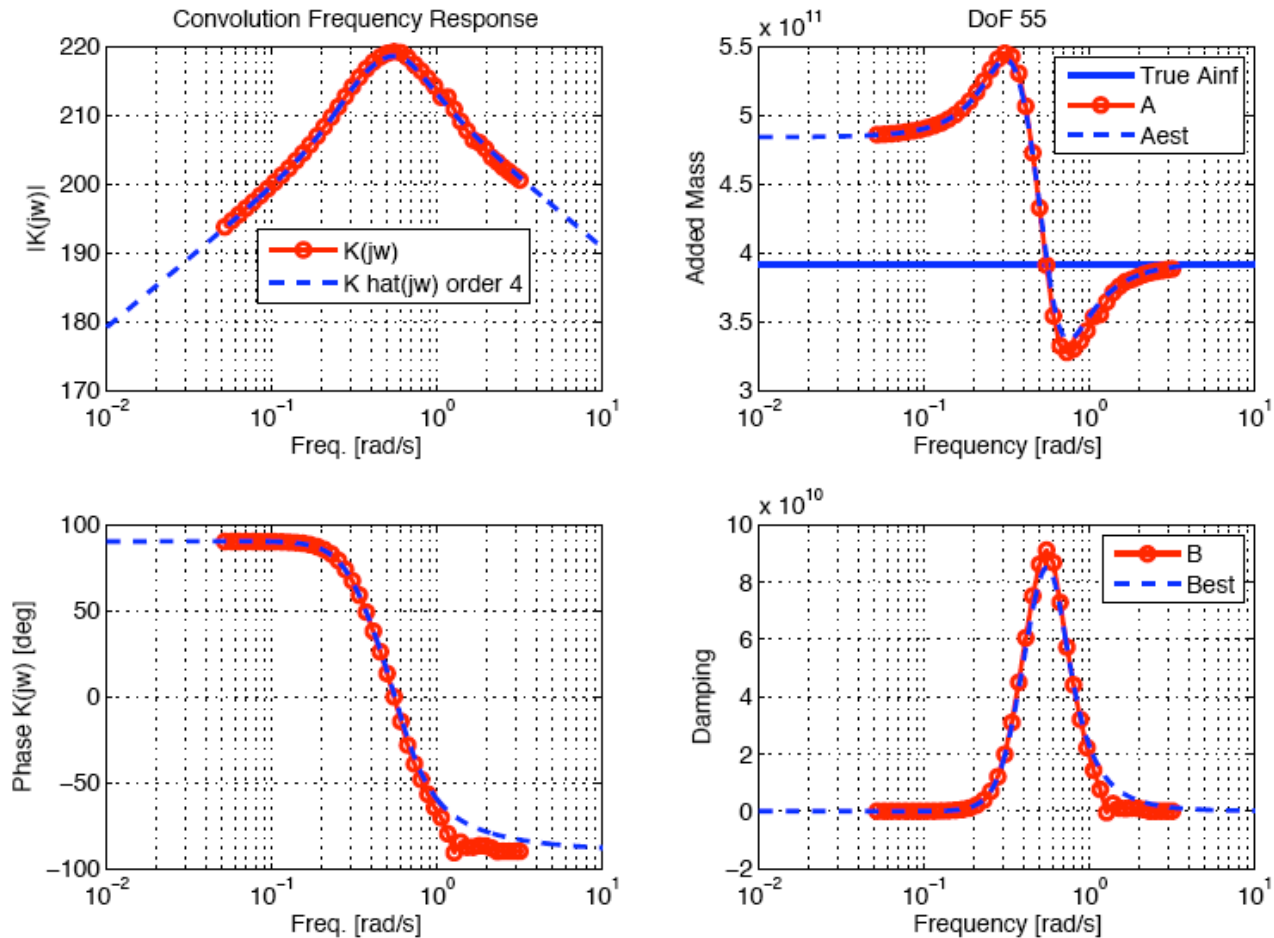
# example FPSO (dof 33)



# example FPSO (dof 35,53)



# example FPSO (dof 55)



# example FPSO

Infinite-frequency added mass coefficients.

| True Value         | Identified               | Rel. Err. |
|--------------------|--------------------------|-----------|
| $A_{33}=1.7283e8$  | $\hat{A}_{33}=1.731e8$   | 1.5 %     |
| $A_{35}=-3.463e7$  | $\hat{A}_{35}=-3.7179e7$ | 0.18%     |
| $A_{55}=3.9154e11$ | $\hat{A}_{55}=3.9293e11$ | 0.35%     |

Similar accuracy is obtained for the other couplings.

# Summary and Conclusions

- We have revisited different methods to the identification of time-domain models based on frequency domain computations.
- Time-domain methods
  - Require forming the impulse response function (distortion).
  - Make no use of prior knowledge (affects model quality).
  - Realization Theory is relatively easy to implement.
  - Applicable to 2D and 3D data.
- Frequency-domain methods
  - Simple to implement and use.
  - Incorporate prior knowledge as constraints (improved model quality).
  - Applicable 2D and 3D data.

# This presentation

- This presentation summarizes the discussions in
  - Perez and Fossen (2008a) ***Time- vs frequency-domain Identification of parametric radiation force models for marine structures at zero speed***. Modeling, Identification and Control Vol. 29 No.1, pp1-19.
  - Perez and Fossen (----) ***Joint Identification of Infinite-frequency Added Mass and Fluid-Memory Models of Marine Structures***. Modeling, Identification and Control. Under review.
  - Perez and Fossen (----) ***Identification of Seakeeping Models from Frequency-response data Enforcing Model structure and parameter Constraints***. To be submitted at Ocean Engineering